Consider Spinal Motion Restriction

Perform Primary Survey
Once Patient has been removed from water

Consider Hypothermia Management

Go to Hypothermia CPG

Respiratory Distress

Administer Oxygen
10-15L/min via NRB mask

EMR

Consider Airway Management

Consider Flowsafe CPAP
5-10cm$H_2$0

EMT-A

If rales are auscultated:

Consider Nebulized Salbutamol
Adults: 5mg
Pediatric: <5years 2.5mg
>5years 5mg

EMT

Cardiac Arrest

Give 5 rescue breaths followed by CPR (30:2) (Adult)

Follow Cardiac Arrest Guidelines

Unnecessary spinal motion restriction can impede adequate opening of the airway and delay delivery of rescue breaths.
Routine spinal motion restriction in the absence of a mechanism that suggests a spinal injury is not recommended

Patient Care Goals

1. Ensure airway, oxygenation, and ventilation as top priorities
2. Treat hypoxia aggressively — it is the primary cause of cardiac arrest in drowning
3. Prevent aspiration-related complications (e.g., pulmonary edema, ARDS)
4. Minimize time to BLS/ALS resuscitation in cardiac arrest
5. Prevent hypothermia and monitor for its progression
6. Provide safe transport and hospital handover, even in asymptomatic patients

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Any patient with a history of water immersion/submersion, including:
  - Unconscious, and/or apneic and/or hypoxic, and/or in cardiac arrest, and/or symptomatic after water submersion.
  - Pediatric or adult drowning events

Exclusion Criteria

- Isolated trauma without water immersion
- Medical cardiac arrest not related to drowning
- Drowning confirmed >20 min submersion with signs of rigor mortis, dependent lividity, or decomposition

Key Patient Assessment Points

1. **Patient History**
   - Time Submerged:
     - Ask or estimate duration of submersion, this is a critical prognostic indicator.
     - Longer submersion (>10 minutes) is associated with worse outcomes
   - Mechanism of Injury:
     - Assess for mechanism which may require spinal motion restriction.
2. **Airway**
   - Is the airway open and protected?
   - Obstruction due to vomit, water, foam, or secretions is common.
   - Listen for gurgling or stridor.
   - Use suction if needed.
3. **Breathing**
   - Assess respiratory rate, effort, and chest rise.
   - Monitor SpO_2, values <94% are abnormal even if the patient appears stable.
   - Auscultate for wheezing, crackles, or decreased air entry; may indicate aspiration or pulmonary edema.
   - Tachypnea or bradypnea indicating fatigue or failure.
4. **Circulation**
   - Check heart rate, blood pressure, capillary refill, and skin color.
   - Bradycardia may be due to hypoxia or hypothermia, especially in children.
   - Palpate central and peripheral pulses.
5. **Disability (Neurological)**
   - Use AVPU or GCS to assess level of consciousness.
   - Check for confusion, lethargy, or unresponsiveness.
   - Monitor for seizures in pediatric cases or post-resuscitation.
6. **Exposure & Environment**
   - Check for hypothermia: cold skin, shivering, or decreased responsiveness. For further managment see Hypothermia CPG.
   - Remove wet clothing and protect from further heat loss.
   - Blood glucose: check in all patients with altered LOC; hypoglycemia may complicate presentation.
7. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

- Red Priority
  - Unresponsive or GCS <9
  - SpO_2 <90% despite oxygen
  - Ongoing respiratory distress or abnormal lung sounds (e.g., crackles, wheezing)
  - Bradycardia, hypotension, or signs of shock
  - ROSC after cardiac arrest
- Yellow Priority
  - Coughing, vomiting, or mild respiratory symptoms with normal vitals
  - Alert and oriented but with history of submersion
  - Minor wheezing or crackles but SpO_2 ≥94% on room air
- Green Priority
  - No symptoms, full recovery on scene, normal exam

Patient Management & Interventions

1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should be routinely requested for:
     - Altered LOC, respiratory distress, or requires advanced airway management.
     - Cardiac arrest present or ROSC achieved
     - Signs of pulmonary edema or suspected aspiration pneumonia
2. **Primary Survey**
   - Perform and Manage Primary Survey as per Universal Care CPG.
   - Airway & Ventilation
   - Suctioning of the airway be required to clear water, vomitus, or foam
   - For unconscious patients, consider airway adjuncts
3. **Administer oxygen**
   - Via NRB or BVM as required.
   - Titrate oxygen to maintain Spo2 >94%
4. **Consider CPAP / BiPaP**
   - Indications:
     - Conscious patient with Moderate to Severe respiratory difficulty.
5. **If No Pulse or No Normal Breathing**
   - Prioritize 5 initial rescue breaths before starting compressions.
   - Perform standard CPR as per Adult or Pediatric Cardiac Arrest CPG.
6. **Spinal Precautions** (Spinal Motion Restriction)
   - Routine Spinal motion restriction is not required.
   - Spinal Motion Restriction is only required if traumatic mechanism (such as shallow diving accident)
7. **Hypothermia**
   - Rewarming if Hypothermic
   - For further management see Hypothermia CPG.
8. **Blood Glucose**
   - Check early in all unconscious or lethargic patients
   - Treat as per Hypoglycemia CPG if glucose low.
9. **Transport**
   - Transport all symptomatic patients, even if improved on scene
   - Prioritize high-risk patients: long submersion, persistent cough, altered LOC, low SpO_2, pediatric.
   - Asymptomatic patients after minor events may be treated at scene with advice to call back or visit emergency department.
10. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.
    - Specific Patient Reassessment Points:

- Airway sounds and breathing effort
- Response to oxygen or airway management
- Signs of fatigue or deterioration en route

Patient Safety Considerations

- Prioritize oxygenation over rapid extrication unless unsafe scene
- Handle hypothermic patients gently to avoid arrhythmias
- Do not delay resuscitation for spinal precautions unless trauma is suspected
- Children may appear stable initially and deteriorate; transport all pediatric cases with symptoms
- Monitor closely for recurrent hypoxia or pulmonary edema

Clinical Insights & Additional Guidance

- Airway and Ventilation Management:
  - Airway Management and ventilation should be early and aggressive.
  - The most serious problem in drowning is lack of oxygen (hypoxia), not the amount of water in the lungs. Even a small amount of water can cause lung irritation and inhibit oxygenation.
  - Do NOT attempt to expel water from lungs (Heimlich/abdominal thrusts), this is ineffective and delays resuscitation.
- Salbutamol
  - Nebulized Salbutamol may be administered if fluid is auscultated, however the priority for management of patients is on the priority measures noted above.
  - Where possible, Salbutamol should be administered via the CPAP nebulizer.
  - The doses of Salbutamol for adults and pediatrics are the same as in asthma.
- Transportation Guidance:
  - Patients may initially recover but develop delayed pulmonary edema or ARDS. Transportation to hospital should be recommended.
  - Always advise caregivers of warning signs for deterioration if not transported.
  - Consult with TeleEMS if clarification is required for borderline or asymptomatic cases
- Pediatrics
  - Pediatric patients have better outcomes even after prolonged submersion; initiate full resuscitation

Documentation

- Identification: Age, sex, weight
- Medical complaint: Drowning or near-drowning
- Information: Submersion time, water type, time found, bystander CPR?
- Signs: RR, SpO_2, HR, BP, GCS, skin color, temperature
- Treatment: Oxygen, suction, airway, CPR, ALS involvement
- Allergies: Known drug or environmental allergies
- Medications: Any meds administered or found at scene
- Background: Medical history, seizures, trauma, cardiac history
- Other: Time of event, ALS/TeleEMS consulted, crew handover

Key Performance Indicators

1. Investigate and document submersion time.
2. Request ALS back-up for red priority patients with a transfer time of >10 minutes.

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